

Philippine Ecosystems, Climate Change, and Overpopulation

The Philippines is one of the world's most biologically diverse countries. Its forests, coral reefs, mangroves, and grasslands shelter thousands of species found nowhere else on Earth. Yet these ecosystems face serious threats. Rising temperatures, changing rainfall patterns, stronger typhoons, and rising sea levels are altering the conditions that life depends on. At the same time, a growing human population places increasing pressure on limited natural resources. Climate change and overpopulation are not separate problems. They interact, amplifying each other's effects on Philippine ecosystems and the organisms that live in them. Understanding these threats — and what can be done about them — is essential for protecting the country's natural heritage.

This reading material describes what is meant by the term "climate change." It explains how biological systems respond to climate change and contribute to understanding its effects on living organisms in the Philippines. It explains how climate change impacts Philippine ecosystems. It proposes ways to adapt to and mitigate the impacts of climate change on local biodiversity. Finally, it analyzes information from secondary sources to evaluate the adverse effects of overpopulation locally and globally.

Part 1: What Is Climate Change?

Definition of Climate Change

Climate change refers to long-term shifts in temperatures and weather patterns. While climate has naturally varied throughout Earth's history, the current period of climate change is driven primarily by human activities, especially the burning of fossil fuels (coal, oil, and natural gas), deforestation, and industrial processes. These activities release **greenhouse gases** such as carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) into the atmosphere, trapping heat and causing global temperatures to rise.

Climate vs. Weather

Feature	Weather	Climate
Time frame	Short-term (hours to days)	Long-term (decades to centuries)
What it describes	Current atmospheric conditions	Average atmospheric conditions over time
Example	"It is raining today."	"The Philippines has a tropical climate with a distinct rainy season."
Variability	Changes rapidly	Changes slowly

Key insight: A single cold day does not disprove climate change. Climate is about long-term trends, not daily fluctuations.

Causes of Climate Change

Cause	Description	Example
Burning of fossil fuels	Releases CO ₂ into the atmosphere	Power plants, vehicles, factories
Deforestation	Removes trees that absorb CO ₂	Clearing of Philippine forests for agriculture and development
Agriculture	Releases methane from livestock and rice paddies	Rice farming produces methane
Industrial processes	Releases various greenhouse gases	Cement production, manufacturing

Waste decomposition Releases methane from landfills Improper waste management

The Greenhouse Effect

The greenhouse effect is a natural process that keeps Earth warm enough to support life. Sunlight passes through the atmosphere and warms Earth's surface, which then radiates heat back toward space. Greenhouse gases such as carbon dioxide, methane, and nitrous oxide trap this heat in the atmosphere and re-emit it back toward Earth. Human activities have intensified this effect by adding excess greenhouse gases, causing more heat to be trapped and leading to global warming.

Step	What Happens
1	Sunlight passes through the atmosphere and warms Earth's surface
2	Earth's surface radiates heat back toward space
3	Greenhouse gases trap some of this heat in the atmosphere
4	Without the greenhouse effect, Earth would be too cold for life
5	With excess greenhouse gases, too much heat is trapped, causing global warming

Evidence of Climate Change

Evidence	Description
Rising global temperatures	Average global temperature has increased by about 1.1°C since pre-industrial times
Melting ice	Glaciers and polar ice sheets are shrinking
Rising sea levels	Thermal expansion and melting ice raise ocean levels

Evidence	Description
Changing rainfall patterns	Some areas experience more drought, others more flooding
More extreme weather	Stronger typhoons, heatwaves, and storms
Ocean acidification	Oceans absorb CO ₂ , becoming more acidic

Part 2: How Biological Systems Respond to Climate Change

Why Biological Systems Are Indicators

Living organisms are sensitive to environmental conditions. Changes in temperature, rainfall, and seasons affect when plants flower, when animals migrate, and where species can survive. Scientists study these responses to understand the effects of climate change on living organisms.

Types of Biological Responses

Response	Description	Example in the Philippines
Phenological shifts	Changes in timing of life events	Earlier flowering of some plant species; earlier arrival of migratory birds
Range shifts	Movement of species to new areas	Some fish species moving to deeper or cooler waters; plants moving to higher elevations
Behavioral changes	Alterations in activity patterns	Animals changing feeding or breeding times
Physiological changes	Changes in body function	Coral bleaching due to warmer water; reduced reproduction in some species

Response	Description	Example in the Philippines
Population changes	Increases or decreases in numbers	Decline of species unable to adapt; increase of invasive species

How Scientists Study These Responses

Method	What It Involves	Example
Long-term monitoring	Observing species over many years	Tracking coral cover on Philippine reefs
Phenological records	Recording timing of seasonal events	Recording when mango trees flower each year
Range mapping	Tracking where species are found	Mapping the spread of invasive species
Remote sensing	Using satellite data to monitor ecosystems	Detecting changes in forest cover
Citizen science	Involving the public in data collection	Reporting sightings of migratory birds

Case Study: Coral Bleaching in the Philippines

Coral reefs are among the most climate-sensitive ecosystems. When water temperatures rise even 1–2°C above normal for extended periods, corals expel the algae (zooxanthellae) that live inside their tissues and provide them with food and color. This is called **coral bleaching**. If temperatures remain high, corals die.

Stage	What Happens
Normal condition	Corals live in symbiosis with algae; reef is colorful and healthy

Stage	What Happens
Heat stress	Water temperature rises; corals become stressed
Bleaching	Corals expel algae; reef turns white
Recovery (possible)	If temperature returns to normal, corals may regain algae
Death (if stress continues)	Corals die; reef ecosystem collapses

Impact on the Philippines: The Philippines lies within the Coral Triangle, the global center of marine biodiversity. Mass bleaching events have damaged reefs in Palawan, Batangas, and other areas, reducing fish populations and threatening food security for coastal communities.

Case Study: Changes in Rice Production

Rice is the staple food of the Philippines. Climate change affects rice farming through changes in rainfall, temperature, and pest patterns.

Climate Factor	Effect on Rice
Higher temperatures	Reduced grain yield; increased sterility
Unpredictable rainfall	Drought stress or flooding
Stronger typhoons	Crop destruction
Rising sea levels	Saltwater intrusion into coastal farms
Changes in pest patterns	New pests; increased infestation

Response: Scientists and farmers are developing heat-tolerant and flood-tolerant rice varieties, adjusting planting calendars, and using water-saving technologies.

Part 3: How Climate Change Impacts Philippine Ecosystems

Major Philippine Ecosystems and Their Vulnerability

Ecosystem	Description	Climate Change Impacts
Tropical rainforests	Dense forests with high biodiversity	Drought stress; increased fire risk; shifts in species composition
Coral reefs	Underwater structures built by corals	Bleaching; reduced calcification; loss of fish habitat
Mangroves	Coastal forests adapted to saltwater	Sea level rise; changes in salinity; storm damage
Seagrass beds	Underwater flowering plants	Reduced light due to sedimentation; temperature stress
Rice terraces	Engineered agricultural landscapes	Changes in rainfall; landslides; erosion
Grasslands and savannas	Open areas with grasses and scattered trees	Drought; fire; shifts in species
Freshwater wetlands	Lakes, rivers, marshes	Changes in water level; warming; invasive species

Specific Impacts on Philippine Ecosystems

1. Coral Reef Degradation

Impact	Explanation
Bleaching	Warmer water causes corals to expel algae and die
Ocean acidification	Lower pH makes it harder for corals to build skeletons
Reduced biodiversity	Loss of habitat for fish and invertebrates
Declining fish catch	Fewer fish for coastal communities

2. Forest Ecosystem Changes

Impact	Explanation
Drought stress	Trees weakened; increased susceptibility to disease
Increased fire risk	Dry conditions make forests more flammable
Species shifts	Plants and animals move to higher elevations
Extinction risk	Species with nowhere to go may disappear

3. Mangrove and Coastal Changes

Impact	Explanation
Sea level rise	Mangroves may be submerged if they cannot migrate inland
Storm damage	Stronger typhoons destroy mangrove forests
Changes in salinity	Affects mangrove growth and reproduction

Impact	Explanation
Loss of coastal protection	Fewer mangroves mean more erosion and storm surge

4. Agricultural Disruption

Impact	Explanation
Reduced crop yields	Heat and drought stress reduce production
Changes in planting seasons	Farmers cannot rely on traditional calendars
Pest and disease outbreaks	Warmer conditions favor some pests
Water scarcity	Irrigation water becomes limited

Feedback Loops

Climate change can create **feedback loops** that amplify its effects.

Feedback Loop	How It Works
Forest dieback	Drought kills trees → fewer trees absorb CO ₂ → more CO ₂ in atmosphere → more warming
Coral mortality	Warm water kills corals → dead reefs cannot support fish → loss of biodiversity and food
Permafrost thaw	(Global) Thawing permafrost releases methane → more warming
Albedo loss	Melting ice exposes darker surfaces → more heat absorbed → more melting

Part 4: Adapting and Mitigating Climate Change Impacts on Biodiversity

Adaptation vs. Mitigation

Term	Definition	Focus
Adaptation	Adjusting to actual or expected climate effects	Reducing vulnerability to impacts
Mitigation	Reducing greenhouse gas emissions or enhancing sinks	Addressing the cause of climate change

Both are needed. Mitigation slows climate change. Adaptation helps ecosystems and communities cope with changes already underway.

Adaptation Strategies for Philippine Biodiversity

Strategy	Description	Example
Protected areas	Establishing and managing conservation areas	Expanding marine protected areas in the Coral Triangle
Coral restoration	Growing and transplanting corals	Coral gardening in Palawan
Mangrove rehabilitation	Replanting and protecting mangroves	Community-based mangrove planting in coastal barangays
Wildlife corridors	Connecting fragmented habitats	Forest corridors linking protected areas

Assisted migration	Moving species to suitable habitats	Relocating heat-sensitive species to cooler areas
Climate-resilient crops	Developing crops tolerant to heat, drought, and floods	Drought-tolerant rice varieties
Early warning systems	Monitoring and predicting climate threats	Typhoon and flood warning systems
Community-based management	Involving local communities in conservation	Community-managed marine sanctuaries

Mitigation Strategies for Philippine Biodiversity

Strategy	Description	Example
Renewable energy	Replacing fossil fuels with clean energy	Solar, wind, and geothermal power
Reforestation	Planting trees to absorb CO ₂	National greening programs
Reducing deforestation	Protecting existing forests	Enforcing logging bans in critical areas
Sustainable agriculture	Farming practices that reduce emissions	Organic farming; reduced rice paddy methane
Waste management	Reducing methane from landfills	Composting; waste segregation
Energy efficiency	Using less energy for the same tasks	LED lighting; efficient appliances
Carbon pricing	Making polluters pay for emissions	Carbon taxes

Individual and Community Actions

Action	Impact
Plant native trees	Absorbs CO ₂ ; provides habitat
Reduce, reuse, recycle	Reduces waste and emissions
Conserve water	Protects freshwater ecosystems
Use public transport or bike	Reduces fossil fuel use
Support local and sustainable products	Reduces carbon footprint
Join community clean-ups	Protects coastal and freshwater ecosystems
Educate others	Builds collective action

Part 5: Overpopulation and Its Effects on Ecosystems

What Is Overpopulation?

Overpopulation occurs when the number of people in an area exceeds the carrying capacity of the environment — that is, when the population cannot be sustained by available resources without degrading the environment. Overpopulation is not simply about the number of people. It is about the relationship between population size, resource use, and environmental impact.

Global Population Trends

Year	Global Population
1800	1 billion
1927	2 billion
1960	3 billion
1974	4 billion
1987	5 billion
1999	6 billion
2011	7 billion
2022	8 billion

The global population has grown rapidly over the past two centuries, driven by advances in medicine, agriculture, and technology that have reduced death rates.

Philippine Population Trends

Year	Philippine Population
1903	7.6 million
1960	27 million
1990	60 million
2010	92 million
2020	109 million

The Philippines is one of the most densely populated countries in Southeast Asia.

Effects of Overpopulation on Ecosystems

Effect	Description	Example in the Philippines
Deforestation	More people need land for housing and farming	Loss of forest cover in Sierra Madre
Habitat destruction	Natural areas converted to human use	Mangrove clearing for fishponds
Overfishing	More people need more fish	Decline of fish stocks in municipal waters
Water scarcity	More people need more water	Groundwater depletion in Metro Manila
Waste accumulation	More people produce more waste	Plastic pollution in Manila Bay
Air and water pollution	More people mean more emissions and sewage	Pollution in Pasig River
Biodiversity loss	Species lose habitat and resources	Endangered species like the Philippine eagle
Climate change acceleration	More people mean more greenhouse gas emissions	Increased carbon footprint

Global Effects of Overpopulation

Effect	Description
Resource depletion	Overuse of fossil fuels, minerals, and water

Effect	Description
Food insecurity	Unable to produce enough food for everyone
Climate change	Higher emissions from more people
Biodiversity loss	Habitat destruction worldwide
Social conflict	Competition for scarce resources
Migration	People move from degraded areas to cities or other countries

Carrying Capacity and Ecological Footprint

Concept	Definition
Carrying capacity	The maximum population an environment can sustain indefinitely
Ecological footprint	The amount of land and resources needed to support a person's lifestyle
Overshoot	When population exceeds carrying capacity

Key insight: The ecological footprint of a population depends not only on its size but also on its consumption patterns. A small population with high consumption can have a larger impact than a large population with low consumption.

Analyzing Information from Secondary Sources

When evaluating the effects of overpopulation, it is important to use credible secondary sources.

Source Type	Examples	What to Look For
Government reports	Philippine Statistics Authority, DENR, NEDA	Population data, resource use, environmental indicators

Source Type	Examples	What to Look For
International organizations	United Nations, World Bank, WWF	Global trends, comparative data
Scientific journals	Peer-reviewed studies	Rigorous methodology, evidence-based conclusions
News reports	Major newspapers, verified online sources	Current events; check for bias
NGO reports	Conservation organizations	Local impacts; community perspectives

Evaluating Sources Critically

Question	Why It Matters
Who produced this information?	Credibility and potential bias
When was it published?	Currency of data
What methodology was used?	Reliability of findings
Is the source peer-reviewed?	Quality control
Does it cite other credible sources?	Supporting evidence
Are there conflicting reports?	Need for balanced analysis

Key Terms

Term	Definition
Climate change	Long-term shifts in temperature and weather patterns, primarily driven by human activities
Greenhouse gases	Gases that trap heat in the atmosphere (CO ₂ , CH ₄ , N ₂ O)
Greenhouse effect	Natural process that warms Earth; intensified by human emissions
Coral bleaching	Loss of algae from corals due to heat stress, causing them to turn white
Phenological shift	Change in the timing of seasonal life events
Range shift	Movement of species to new geographic areas
Adaptation	Adjusting to climate change impacts
Mitigation	Reducing greenhouse gas emissions
Overpopulation	When population exceeds the carrying capacity of the environment
Carrying capacity	Maximum population an environment can sustain
Ecological footprint	Land and resources needed to support a lifestyle
Biodiversity	The variety of life in a given area

Review Questions

1. Define climate change. Differentiate it from weather.
2. List four causes of climate change and explain how each contributes to global warming.
3. Describe three ways biological systems respond to climate change. Provide one Philippine example for each.
4. Explain how coral bleaching occurs and why it is a serious threat to Philippine marine ecosystems.
5. Identify four Philippine ecosystems and describe one climate change impact on each.
6. Differentiate adaptation from mitigation. Provide two examples of each for Philippine biodiversity.
7. Propose three ways individuals and communities can help mitigate climate change impacts on local biodiversity.
8. Define overpopulation. Explain why it is not simply about the number of people.
9. Study the data below and answer the questions that follow.

Year	Philippine Population
1990	60 million
2000	76 million
2010	92 million
2020	109 million

- (a) Describe the trend in Philippine population from 1990 to 2020.
- (b) Identify three environmental effects that could result from this trend.
- (c) Explain how overpopulation and climate change can interact to worsen environmental problems.

10. Application question: A coastal barangay in the Philippines depends on fishing and farming for its livelihood. In recent years, residents have noticed fewer fish, damaged coral reefs, unpredictable rainfall, and saltwater intrusion into rice fields. The population of the barangay has also grown.

- (a) Identify the climate-related and population-related factors affecting the barangay.
- (b) Explain how these factors interact to threaten local biodiversity and food security.
- (c) Propose three adaptation strategies and two mitigation strategies the community could adopt.
- (d) Explain how secondary sources could help the community evaluate the severity of the problem and plan appropriate responses.